CPSC6440 Problem Set 1

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Question 2.1.

Proof. By definition,

$$P_{ij} = (D^{-1}W)_{ij} = \frac{W_{ij}}{d_i}, \qquad d_i = \sum_{j=1}^{n} W_{ij}.$$

Hence each row of P sums to one:

$$(P\mathbf{1})_i = \sum_{j=1}^{n} P_{ij} \cdot 1 = \sum_{j=1}^{n} P_{ij} = 1 \quad \Longrightarrow \quad P\mathbf{1} = \mathbf{1}.$$

Thus $\mathbf{1}$ is an eigenvector of P with associated eigenvalue $\lambda = 1$. □

Question 2.2.

Proof. P and P_s are similar ($P_s = D^{1/2}PD^{-1/2}$), hence $\lambda_i(P) = \lambda_i(P_s)$.

Since P_s is real symmetric, the spectral theorem applies: $\exists$ an orthogonal matrix Q and a real diagonal matrix Λ such that

$$P_s = Q\Lambda Q^\top.$$

Therefore all eigenvalues of P_s (and hence of P) are real.

Substitution gives:

$$P = D^{-1/2}Q\Lambda Q^\top D^{1/2}.$$

Thus P is diagonalizable over $\mathbb{R}$ with decomposition

$$P = U\Lambda U^{-1}, \qquad U = D^{-1/2}Q, \quad U^{-1} = Q^\top D^{1/2}.$$

□

Question 2.3.

Proof. By spectral decomposition, the t-step transition probabilities from x to z is:

$$p_t(x,z) \;=\; (P^t)(x,z) \;=\; \sum_{i=1}^{n} \lambda_i^t \psi_i(x)\psi_i(z)\phi_0(z)$$

where:

- λ_i are the eigenvalues
- ψ_i are the right eigenvectors
- ϕ_0 is the stationary distribution of the reversible Markov chain

For two points x and y, the different in transition probabilities to any z is:

$$p_t(x,z) - p_t(y,z) \;=\; \phi_0(z) \sum_{i=1}^{n} \lambda_i^t \big(\psi_i(x) - \psi_i(y)\big)\psi_i(z),$$

Now compute the diffusion distance:

$$\begin{aligned} D_t^2(x,y) &= \sum_{z\in\mathcal{X}} \frac{\big(p_t(x,z) - p_t(y,z)\big)^2}{\phi_0(z)} \\ &= \sum_{z\in\mathcal{X}} \frac{1}{\phi_0(z)} \left(\phi_0(z) \sum_{i=1}^{n} \lambda_i^t (\psi_i(x) - \psi_i(y))\psi_i(z) \right)^2 \\ &= \sum_{z\in\mathcal{X}} \phi_0(z) \left(\sum_{i=1}^{n} \lambda_i^t (\psi_i(x) - \psi_i(y))\psi_i(z) \right)^2 \end{aligned}$$

By expanding the square:

$$\begin{aligned} D_t^2(x,y) &= \sum_{z\in\mathcal{X}} \phi_0(z) \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i^t \lambda_j^t (\psi_i(x) - \psi_i(y))(\psi_j(x) - \psi_j(y))\psi_i(z)\psi_j(z) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i^t \lambda_j^t (\psi_i(x) - \psi_i(y))(\psi_j(x) - \psi_j(y)) \sum_{z\in\mathcal{X}} \phi_0(z)\psi_i(z)\psi_j(z) \end{aligned}$$

Using orthonormality

$$\sum_{z\in\mathcal{X}} \phi_0(z)\psi_i(z)\psi_j(z) = \delta_{ij},$$

we obtain

$$D_t^2(x,y) = \sum_{i=1}^{n} \lambda_i^{2t} \big(\psi_i(x) - \psi_i(y)\big)^2.$$

The RHS is the squared Euclidean distance between the embeddings $\Psi_t(x)$ and $\Psi_t(y)$:

$$\|\Psi_t(x) - \Psi_t(y)\|^2 = \sum_{i=1}^{n} \big(\lambda_i^t \psi_i(x) - \lambda_i^t \psi_i(y)\big)^2.$$

□

Question 3.1

NOTE: I have spoken to the TA about defining my own plotting functions for 2D and 3D so I can make subplots, add color bar, and generate the 3D plot interactively.

1. How do the Diffusion Map visualizations look?
   The 2D and 3D visualizations show a curved trajectory structure. The color gradient (representing time steps) flows along the branch. This flow is especially clear in the 2D components 2 vs 3 plot.

2. Are they capturing the time progression of the data set?
   Yes. The color gradient shows a transition from purple (early time points) to bright yellow (late time points). Early time points (purple/dark blue) cluster together at one end of the trajectory. Late time points (yellow) are at the opposite end. The progression indicates that the diffusion map preserved the temporal ordering.

3. Do you see any clusters forming?
   Yes.

   - Early stage cluster: Dense grouping of purple points (time steps 0-500)
   - Intermediate stage cluster: Gradual transition through the branches (dark blue and green points, time steps 500-1500)
   - Late stage cluster: Light green and yellow points at the ends of the branches (time steps 1500-2000)

4. What is the intrinsic dimensionality given by Diffusion Maps?

   The intrinsic dimensionality is 3 dimensions. From the eigenvalue plot, we see the first eigenvalue = 1 as it should be, the second eigenvalue ≈ 0.8, and the third eigenvalue ≈ 0.75, then a large gap between the third and forth eigenvalue. This drop suggests that there's good connectivity along the first 3 diffusion directions which capture the essential geometry of how cells transition through reprogramming space.

5. Which channels are most highly correlated with diffusion components?
   — Top 5 channels for Diffusion Component 1:

- Channel: h3k9ac, Absolute Correlation: 0.8168
- Channel: prb, Absolute Correlation: 0.7960
- Channel: pplk1, Absolute Correlation: 0.7462
- Channel: pstat3-727, Absolute Correlation: 0.6688
- Channel: pampk, Absolute Correlation: 0.6393

— Top 5 channels for Diffusion Component 2:

- Channel: oct4, Absolute Correlation: 0.7384
- Channel: mefsk4, Absolute Correlation: 0.6265
- Channel: cd140a, Absolute Correlation: 0.6171
- Channel: bcatenin, Absolute Correlation: 0.5985
- Channel: thy1, Absolute Correlation: 0.5020

— Top 5 channels for Diffusion Component 3:

- Channel: lin28, Absolute Correlation: 0.5968
- Channel: epcam, Absolute Correlation: 0.5815
- Channel: ssea1, Absolute Correlation: 0.5713
- Channel: cd44, Absolute Correlation: 0.4389
- Channel: mefsk4, Absolute Correlation: 0.4189

6. Any guesses to the biological interpretation?

- Component 1 seems to capture the cell cycle and proliferation changes during reprogramming (H3K9ac, PLK1, Rb, STAT3, AMPK are all cell cycle/growth related)
- Component 2 seems to represent pluripotency acquisition (OCT4 is one of the four "Yamanaka factors," a set of transcription factors used by Shinya Yamanaka to reprogram adult somatic cells into iPSCs; β-catenin is also crucial for inducing pluripotency)
- Component 3 seems to also capture pluripotency and cell adhesion(lin28, EpCAM, and SSEA1 are related to pluripotency; CD44 and EpCAM are involved in cell adhesion)

Question 3.2

1. Did the choice of t affect the embeddings and, if so, how? Do the corresponding eigenvalues support your findings?

   Yes, as we increase t, the axes of the the plots change, showing more compressed trajectory with less fine-grained temporal details. The eigenvalue plots are identical which is expected since t only affects the final embedding by weighting eigenvectors with eigenvalue^t. Larger t suppresses higher-order eigenvectors but do not change the eigenvalues themselves.

2. How did the choice of k affect the embeddings and, if so, how?

   Yes. When $k = 5$, the trajectory appears more compact while when $k = 8$, the trajectory is more spread out and shows different geometric structure this is because smaller k creates more local neighborhoods than larger k. In the eigenvalue plots, we see more rapid drop when $k = 8$.

3. Did previously observed trends (i.e. clusters, time progression) in diffusion dimensions change dramatically?

   No. The color gradient remained consistent across parameter choices, suggesting that the biological trajectory seems robust. Though with higher t, more smoothing is applied, and with different k, cluster boundaries and shapes changed.

4. Did the correlated channels change with σ? How would you interpret this in terms of the data?

   Yes, the correlated channels changed with different parameters; however, some key channels such as h3k9ac, oct4, and thy1, robustly representing biological processes essential to reprogramming remained consistently important. In terms of the data, these changes suggest that the reprogramming process has multi-scale structure. Some biological signals might only become visible at certain levels of analysis resolution; therefore, parameter-sensitive markers may represent scale-dependent processes.

Question 3.3

1. How do the PHATE and tSNE visualizations compare to the diffusion maps visualization?

   PHATE shows a more complex, branched structure with multiple branches and bifurcations rather than a single smooth curve. It maintains some continuity and preserves local neighborhoods, but has more detailed local structure compared to diffusion maps.

   tSNE shows discrete, separated clusters rather than a continuous trajectory. It breaks the manifold into distinct groups. Global structure is lost as there is no clear sense of the overall trajectory shape. It emphasizes local clustering at the expense of global connectivity. The reprogramming process appears as discrete cell states rather than a continuous process.

2. Are they capturing the time progression of the data set?

   PHATE - Yes, time still flows in a logical direction, but shows that reprogramming might have multiple routes to the same endpoint.

   tSNE- No, early and late time points are mixed. No clear directional flow of time progression. Time points appear randomly scattered across clusters.

3. Do you see any clusters forming?

   Diffusion maps and PHATE both show temporal clusters, but PHATE seems to also show some intermediate clustering. tSNE clusters seem to have artificial boundaries rather than representing biological clusters as the temporal trajectory is broken.

Question 3.4

1. How do the 3 different clustering methods compare (Louvain, K-means, Spectral Clutering)?

   Louvain shows clear progression along the temporal trajectory with distinct but connected communities. Both K-means and spectral clustering broke the temporal continuity. The three algorithms are fundamentally different. Louvain clustering is about finding communities within a connected network. K-means is about finding regions that minimize distances to centroids. Spectral clustering is about optimally partitioning a graph into disconnected components

2. Which is following the shape of the data the best?

   Louvain. With 12 clusters found, it captures the natural developmental process.

3. Does one of the clustering methods "match" well with any of the visualizations?

   Louvain clustering aligns best with what we see in the diffusion map visualizations. The clusters form a continuous progression from one end of the trajectory to the other.

Question 3.5

Derive the incremental update to the modularity equation when a point moves from one module i to another j. The Louvain algorithm optimizes the modularity function defined in Equation 1.

I was a bit confused by the instructions as i and j are nodes not communities while Q is calculated over communities. After clarifying with the TA, the following derivation was given assuming that "when a point moves from one module i to another j" meant when a node i moves from one community d to another community c.

$$Q = \frac{1}{2m} \sum_{(i,j) \in E} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{1}$$

Equation 1 can also be written as:

$$Q = \sum_{c \in \Gamma} \left[\frac{\sigma_c}{2m} - \left(\frac{\Sigma_c}{2m} \right)^2 \right] \tag{2}$$

Where:

- σ_c = sum of weights of edges within community c
- Σ_c = sum of degrees of all nodes in community c
- Γ = set of all communities

When moving a node i from community d to c, the change in modularity is:

$$\Delta Q = \Delta Q_d \ + \Delta Q_c$$

As community d loses node i, it also loses all its internal edges. Hence:

$$\Delta Q_d = \left[\frac{\sigma_d - k_{i \to d}}{2m} - \left(\frac{\Sigma_d - k_i}{2m} \right)^2 \right] - \left[\frac{\sigma_d}{2m} - \left(\frac{\Sigma_d}{2m} \right)^2 \right] \tag{3}$$

$$= \frac{\Sigma_d k_i}{2m^2} - \frac{k_{i \to d}}{2m} - \frac{k_i^2}{4m^2} \tag{4}$$

Similarly, as community c gains node i, in also gains edges. Hence:

$$\Delta Q_c = \left[\frac{\sigma_c + k_{i\to c}}{2m} - \left(\frac{\Sigma_c + k_i}{2m}\right)^2\right] - \left[\frac{\sigma_c}{2m} - \left(\frac{\Sigma_c}{2m}\right)^2\right] \tag{5}$$

$$= -\frac{\Sigma_c k_i}{2m^2} + \frac{k_{i\to c}}{2m} - \frac{k_i^2}{4m^2} \tag{6}$$

Plugging 4 and 6 into 2 gives:

$$\begin{aligned}
\Delta Q &= \frac{\Sigma_d k_i}{2m^2} - \frac{k_{i\to d}}{2m} - \frac{k_i^2}{4m^2} - \frac{\Sigma_c k_i}{2m^2} + \frac{k_{i\to c}}{2m} - \frac{k_i^2}{4m^2} \\
&= \frac{1}{2m}(k_{i\to c} - k_{i\to d}) + \frac{k_i}{2m^2}(\Sigma_d - \Sigma_c) - \frac{k_i^2}{2m^2} \\
&= \frac{1}{2m}(k_{i\to c} - k_{i\to d}) - \frac{k_i}{2m^2}(\Sigma_c - \Sigma_d + k_i)
\end{aligned}$$

References

[1] Coifman, R. R., and Lafon, S. Diffusion maps. *Applied and Computational Harmonic Analysis 21*, 1 (jul 2006), 5–30.

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[3] Zunder, E., Lujan, E., Goltsev, Y., Wernig, M., and Nolan, G. A Continuous Molecular Roadmap to iPSC Reprogramming through Progression Analysis of Single-Cell Mass Cytometry. *Cell Stem Cell 16*, 3 (mar 2015), 323–337.